

AMNA BIBI

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PERSONAL STATEMENT

A motivated Computer Engineering student with strong programming and analytical skills in Python, Java & C++. Responsible, detail-oriented, and consistent in delivering quality work. Eager to apply technical knowledge in practical projects and continue growing professionally.

EDUCATION

National University of Sciences & Technology (NUST)

Islamabad, Pakistan

Bachelor of Engineering – Computer Engineering

Sep 2024 – PRESENT

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems, Object-Oriented Programming, Software Engineering, Computer Networks

PROJECTS

Reinforcement Learning-Based Bus Control Algorithm (In Progress)

- Designing a Deep Reinforcement Learning (DQN/PPO) based controller for multi-bus communication systems
- Modeling the system as a Markov Decision Process (MDP) with defined state, action, and reward spaces
- Developing a simulation to optimize latency, jitter, and packet loss

Autonomous Drone Fleet Coordination Simulator

Academic Project — 2025

- Developed a multi-agent drone simulation using Object-Oriented Programming (OOP) in Java
- Modelled drone dynamics using Newton–Euler equations for realistic motion and control
- Implemented modules for collision avoidance, formation control, and inter-drone communication
- Designed a modular architecture including Simulator, Drone, Controller, and Environment classes
- Simulated real-world constraints such as communication range, stochastic data exchange, and boundary conditions

Minesweeper Game (Console-Based)

Academic Project — 2024

- Developed a console-based Minesweeper game using programming fundamentals
- Applied recursive logic for revealing adjacent empty cells
- Included basic win/loss conditions and user input handling

SKILLS

Programming Languages: Python, Java, C++, C

Core Concepts: Object-Oriented Programming (OOP), Data Structures & Algorithms, Problem Solving

Web Technologies: HTML, CSS

Tools & Technologies: File Handling, Simulation Development, GUI, CSV/Data Logging, Debugging, Visual Studio, Visual Studio Code, System Verilog, LTSpice, Proteus.

Operating Systems: Windows

SOFT SKILLS

Time management, Team Collaboration, Project Management & Adaptability



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Jazz
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Subject: **Internship Request for Amna Bibi**

Sir,

I take this opportunity to acknowledge the contribution of your esteemed organization in national development and economic uplift of the country. I would also like to apprise you of the fact that College of Electrical and Mechanical Engineering, a constituent college of NUST in its bid to provide quality technical education tailored to match the local industrial/organizational requirements, made it compulsory for every student to undergo an internship of 6-8 weeks with some reputed organization to link theoretical knowledge to practical situations.

We appreciate that this on-the-job training is not possible without a strong commitment from private sector / industrial organizations. Since it's a matter of national importance, needless to say that we shall do all possible within our means and expect organizations like yours to reciprocate and contribute equally. This will not only groom technocrats at national level but also add potential engineers at your disposal.

Amna Bibi is a student of **Computer Engineering** department of this college. She is an extremely talented and gifted student who displayed tremendous talents and abilities in academics and extra-curricular alike. It is requested to accommodate her as an internee for a period of 6-8 weeks. We look forward for a positive response.

Thanking you in anticipation.

Dated: 22 April 2026.

ENGR. FARHAN ASLAM KHAN
Industry Liaison Officer (CME)
National University of Sciences & Technology
(NUST), Islamabad. 051-54444704

Engr. Farhan Aslam Khan
(Industry Liaison Officer, NUST)